How to Install Kubernetes Cluster on Ubuntu 22.04

Last Updated: July 23, 2023 by Pradeep Kumar

AD

Are you looking for an easy guide on how to install Kubernetes Cluster on Ubuntu 22.04 (Jammy Jellyfish)?

The step-by-step guide on this page will show you how to install Kubernetes cluster on Ubuntu 22.04 using Kubeadm command step by step.

Kubernetes has become the de facto container orchestration platform, empowering developers and system administrators to manage and scale containerized applications effortlessly. If you're an Ubuntu 22.04 user eager to harness the power of Kubernetes, you've come to the right place.

A Kubernetes cluster consists of worker nodes on which application workload is deployed and a set up master nodes which are used to manage worker nodes and pods in the cluster.

Prerequisites

In this guide, we are using one master node and two worker nodes. Following are system requirements on each node,

- Minimal install Ubuntu 22.04
- Minimum 2GB RAM or more
- Minimum 2 CPU cores / or 2 vCPU
- 20 GB free disk space on /var or more
- Sudo user with admin rights
- Internet connectivity on each node

Lab Setup

- Master Node: 192.168.1.173 – k8smaster.example.net
- First Worker Node: 192.168.1.174 – k8sworker1.example.net
- Second Worker Node: 192.168.1.175 – k8sworker2.example.net

Without any delay, let's jump into the installation steps of Kubernetes cluster

1) Set hostname on Each Node

Login to to master node and set hostname using hostnamectl command,

```
$ sudo hostnamectl set-hostname "k8smaster.example.net"
$ exec bash
```

On the worker nodes, run

```
$ sudo hostnamectl set-hostname "k8sworker1.example.net"   // 1st worker node
$ sudo hostnamectl set-hostname "k8sworker2.example.net"   // 2nd worker node
$ exec bash
```

Add the following entries in /etc/hosts file on each node

```
192.168.1.173   k8smaster.example.net k8smaster
192.168.1.174   k8sworker1.example.net k8sworker1
192.168.1.175   k8sworker2.example.net k8sworker2
```

2) Disable swap & Add kernel Parameters

Execute beneath swapoff and sed command to disable swap. Make sure to run the following commands on all the nodes.

```
$ sudo swapoff -a
$ sudo sed -i '/ swap / s/^\(.*\)$/#\1/g' /etc/fstab
```

Load the following kernel modules on all the nodes,

```
$ sudo tee /etc/modules-load.d/containerd.conf <<EOF
overlay
br_netfilter
EOF
$ sudo modprobe overlay
$ sudo modprobe br_netfilter
```

Set the following Kernel parameters for Kubernetes, run beneath tee command

```
$ sudo tee /etc/sysctl.d/kubernetes.conf <<EOF
net.bridge.bridge-nf-call-ip6tables = 1
net.bridge.bridge-nf-call-iptables = 1
net.ipv4.ip_forward = 1
EOF
```

Reload the above changes, run

```
$ sudo sysctl --system
```

3) Install Containerd Runtime

In this guide, we are using containerd runtime for our Kubernetes cluster. So, to install containerd, first install its dependencies.

```
$ sudo apt install -y curl gnupg2 software-properties-common apt-transport-ht
```

Enable docker repository

```
$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --d
$ sudo add-apt-repository "deb [arch=amd64] https://download.docker.com/linux
```

Now, run following apt command to install containerd

```
$ sudo apt update
$ sudo apt install -y containerd.io
```

Configure containerd so that it starts using systemd as cgroup.

```
$ containerd config default | sudo tee /etc/containerd/config.toml >/dev/null
$ sudo sed -i 's/SystemdCgroup \= false/SystemdCgroup \= true/g' /etc/contain
```

Restart and enable containerd service

```
$ sudo systemctl restart containerd
$ sudo systemctl enable containerd
```

4) Add Apt Repository for Kubernetes

Kubernetes package is not available in the default Ubuntu 22.04 package repositories. So we need to add kubernetes repositories, run following command,

```
$ curl -s https://packages.cloud.google.com/apt/doc/apt-key.gpg | sudo gpg --
$ sudo apt-add-repository "deb http://apt.kubernetes.io/ kubernetes-xenial ma
```

Note: At time of writing this guide, Xenial is the latest Kubernetes repository but when repository is available for Ubuntu 22.04 (Jammy Jellyfish) then you need replace xenial word with 'jammy' in 'apt-add-repository' command.

5) Install Kubectl, Kubeadm and Kubelet

Post adding the repositories, install Kubernetes components like kubectl, kubelet and Kubeadm utility on all the nodes. Execute following set of commands,

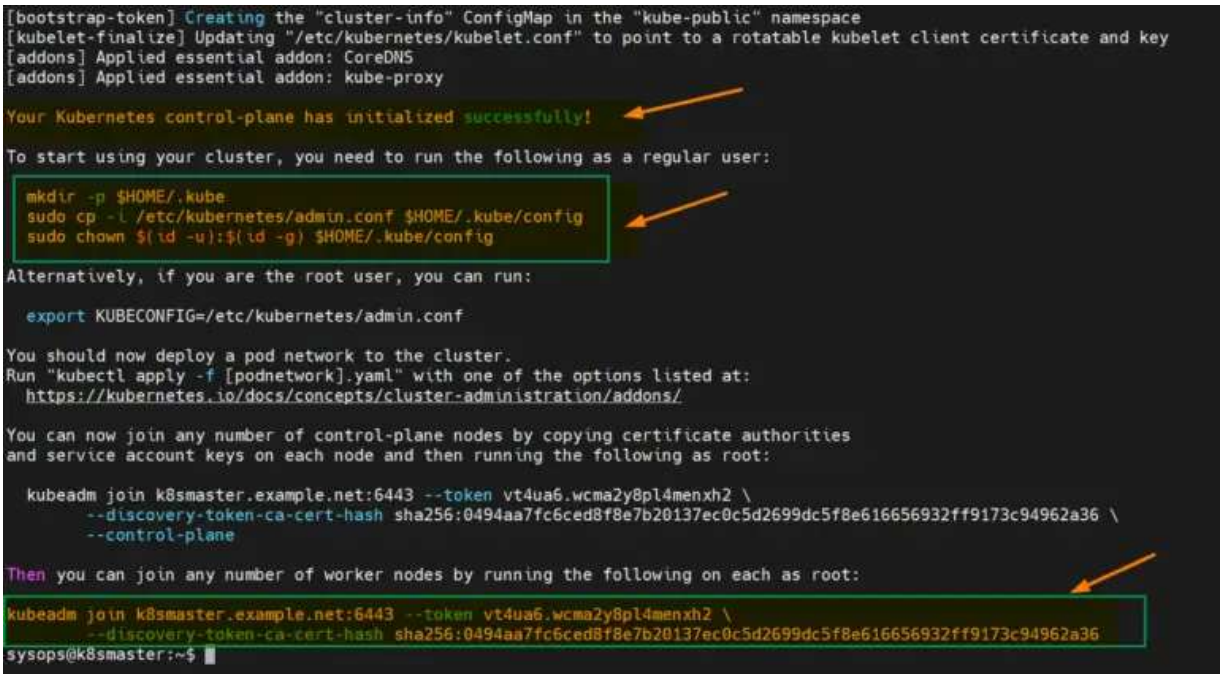
```
$ sudo apt update
$ sudo apt install -y kubelet kubeadm kubectl
$ sudo apt-mark hold kubelet kubeadm kubectl
```

6) Initialize Kubernetes Cluster with Kubeadm

Now, we are all set to initialize Kubernetes cluster. Run the following Kubeadm command on the master node only.

```
$ sudo kubeadm init --control-plane-endpoint=k8smaster.example.net
```

Output of above command,



After the initialization is complete, you will see a message with instructions on how to join worker nodes to the cluster. Make a note of the kubeadm join command for future reference.

So, to start interacting with cluster, run following commands on the master node,

```
$ mkdir -p $HOME/.kube
$ sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
$ sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

next, try to run following kubectl commands to view cluster and node status

```
$ kubectl cluster-info
$ kubectl get nodes
```

Output,

```
sysops@k8smaster:~$ kubectl cluster-info
Kubernetes control plane is running at https://k8smaster.example.net:6443
CoreDNS is running at https://k8smaster.example.net:6443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
sysops@k8smaster:~$ kubectl get nodes
NAME        STATUS     ROLES           AGE     VERSION
k8smaster   NotReady   control-plane   3m51s   v1.24.3
sysops@k8smaster:~$
```

7) Join Worker Nodes to the Cluster

On each worker node, use the kubeadm join command you noted down earlier after initializing the master node on step 6. It should look something like this:

```
$ sudo kubeadm join k8smaster.example.net:6443 --token vt4ua6.wcma2y8pl4menxh
   --discovery-token-ca-cert-hash sha256:0494aa7fc6ced8f8e7b20137ec0c5d2699dc
```

Output from both the worker nodes,

```
sysops@k8sworker1:~$ sudo kubeadm join k8smaster.example.net:6443 --token vt4ua6.wcma2y8pl4menxh2 \
        --discovery-token-ca-cert-hash sha256:0494aa7fc6ced8f8e7b20137ec0c5d2699dc5f8e616656932ff9173c94962a36
[sudo] password for sysops:
[preflight] Running pre-flight checks
        [WARNING SystemVerification]: missing optional cgroups: blkio
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubelet-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubelet-start] Starting the kubelet
[kubelet-start] Waiting for the kubelet to perform the TLS Bootstrap...

This node has joined the cluster:
* Certificate signing request was sent to apiserver and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.

sysops@k8sworker1:~$
```

```
sysops@k8sworker2:~$ sudo kubeadm join k8smaster.example.net:6443 --token vt4ua6.wcma2y8pl4menxh2 \
        --discovery-token-ca-cert-hash sha256:0494aa7fc6ced8f8e7b20137ec0c5d2699dc5f8e616656932ff9173c94962a36
[sudo] password for sysops:
[preflight] Running pre-flight checks
        [WARNING SystemVerification]: missing optional cgroups: blkio
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubelet-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubelet-start] Starting the kubelet
[kubelet-start] Waiting for the kubelet to perform the TLS Bootstrap...

This node has joined the cluster:
* Certificate signing request was sent to apiserver and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.

sysops@k8sworker2:~$
```

Above output from worker nodes confirms that both the nodes have joined the cluster.Check the nodes status from master node using kubectl command,

```
$ kubectl get nodes
```

```
sysops@k8smaster:~$ kubectl get nodes
NAME                    STATUS     ROLES           AGE     VERSION
k8smaster               NotReady   control-plane   27m     v1.24.3
k8sworker1.example.net  NotReady   <none>          6m3s    v1.24.3
k8sworker2.example.net  NotReady   <none>          5m39s   v1.24.3
sysops@k8smaster:~$
```

As we can see nodes status is 'NotReady', so to make it active. We must install CNI (Container Network Interface) or network add-on plugins like Calico, Flannel and Weave-net.

8) Install Calico Network Plugin

A network plugin is required to enable communication between pods in the cluster. Run following kubectl command to install Calico network plugin from the master node,

```
$ kubectl apply -f https://raw.githubusercontent.com/projectcalico/calico/v3.
```

Output of above commands would look like below,

```
sysops@k8smaster:~$ curl https://projectcalico.docs.tigera.io/manifests/calico.yaml -O
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  225k  100  225k    0     0   126k      0  0:00:01  0:00:01 --:--:--  145k
sysops@k8smaster:~$ kubectl apply -f calico.yaml
configmap/calico-config created
customresourcedefinition.apiextensions.k8s.io/bgpconfigurations.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/bgppeers.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/blockaffinities.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/caliconodestatuses.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/clusterinformations.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/felixconfigurations.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/globalnetworkpolicies.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/globalnetworksets.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/hostendpoints.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/ipamblocks.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/ipamconfigs.crd.projectcalico.org created
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customresourcedefinition.apiextensions.k8s.io/networkpolicies.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/networksets.crd.projectcalico.org created
clusterrole.rbac.authorization.k8s.io/calico-kube-controllers created
clusterrolebinding.rbac.authorization.k8s.io/calico-kube-controllers created
clusterrole.rbac.authorization.k8s.io/calico-node created
clusterrolebinding.rbac.authorization.k8s.io/calico-node created
daemonset.apps/calico-node created
serviceaccount/calico-node created
deployment.apps/calico-kube-controllers created
serviceaccount/calico-kube-controllers created
poddisruptionbudget.policy/calico-kube-controllers created
sysops@k8smaster:~$
```

Verify the status of pods in kube-system namespace,

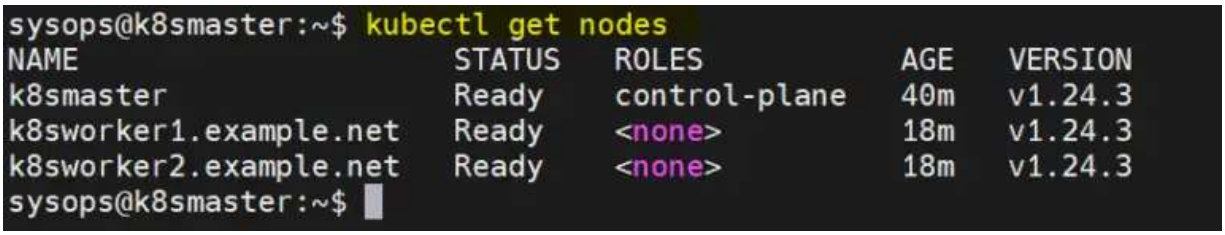
```
$ kubectl get pods -n kube-system
```

Output,



Perfect, check the nodes status as well.

```
$ kubectl get nodes
```



Great, above confirms that nodes are active node. Now, we can say that our Kubernetes cluster is functional.

9) Test Your Kubernetes Cluster Installation

To test Kubernetes installation, let's try to deploy nginx based application and try to access it.

```
$ kubectl create deployment nginx-app --image=nginx --replicas=2
```

Check the status of nginx-app deployment

```
$ kubectl get deployment nginx-app
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
nginx-app   2/2     2            2           68s
$
```

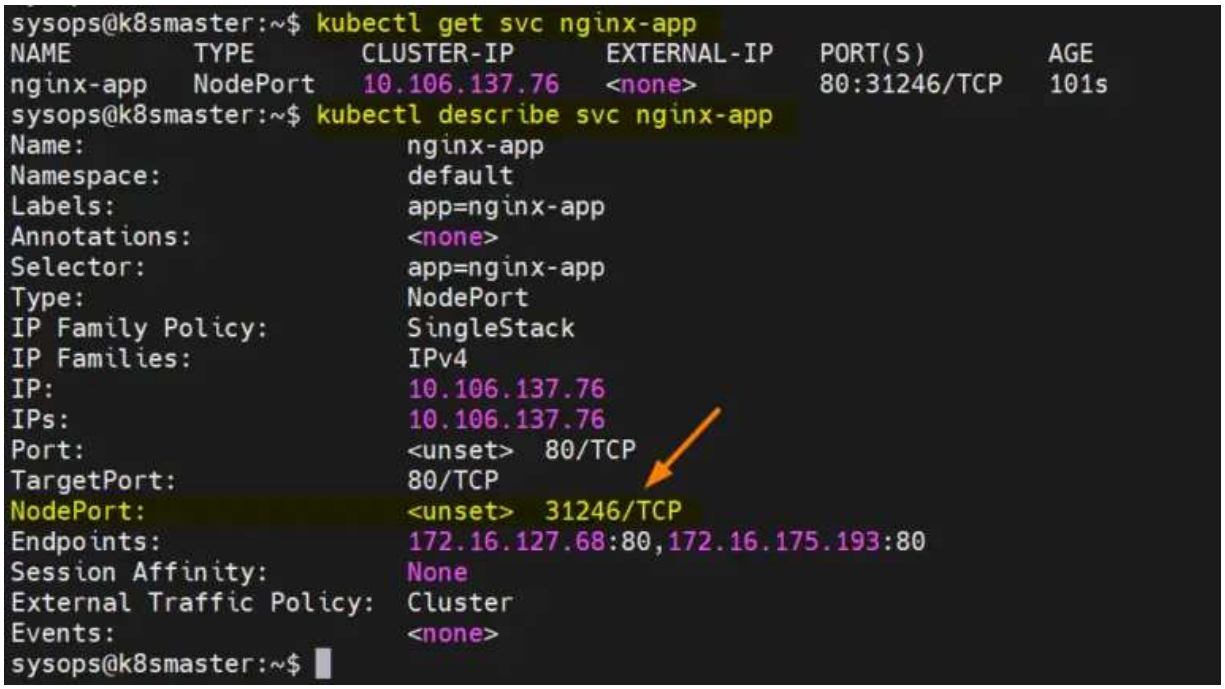
Expose the deployment as NodePort,

```
$ kubectl expose deployment nginx-app --type=NodePort --port=80
service/nginx-app exposed
$
```

Run following commands to view service status

```
$ kubectl get svc nginx-app
$ kubectl describe svc nginx-app
```

Output of above commands,



Use following curl command to access nginx based application,

$ curl http://<woker-node-ip-addres>:31246

```
$ curl http://192.168.1.174:31246
```

Output,

```
sysops@k8smaster:~$ curl http://192.168.1.174:31246
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
sysops@k8smaster:~$
```

Great, above output confirms that nginx based application is accessible.

That's all from this guide, I hope you have found this guide useful. Kindly do post your queries and feedback in below comments section.

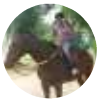
Also Read: How to Install and Access Kubernetes Dashboard Step-by-Step

Conclusion:

Congratulations! You have successfully set up a Kubernetes cluster on Ubuntu 22.04. With Kubernetes at your disposal, you can now orchestrate, scale, and manage your containerized applications efficiently. Explore further by deploying more complex applications and services on your Kubernetes cluster, and take full advantage of the power and flexibility it offers.

Also Read: How to Install Kubernetes (K8s) Metrics Server Step by Step

63 thoughts on "How to Install Kubernetes Cluster on Ubuntu 22.04"



Ashish
August 16, 2022 at 7:15 pm

after going through 100s of videos and different blog post , finally your document helped me to setup working kubernetes cluster….kudoz

Reply



Pradeep Kumar
August 17, 2022 at 1:39 pm

Thanks Ashish !! for your feedback.

I am glad, this post helps you to deploy Kubernetes Cluster.

Reply



Niels Basjes
August 26, 2022 at 3:13 pm

The commands adding keys for the apt repos should be changed to something like this:

sudo curl -fsSL ‘https://download.docker.com/linux/ubuntu/gpg’ | sudo gpg –dearmour -o /etc/apt/trusted.gpg.d/docker.gpg

Reply



Pradeep Kumar
August 27, 2022 at 2:54 am

Hi Niels,

Thanks for sharing the updated command. As per your suggestion, I have modified command in article as well.

Reply



Carlos Tous
September 4, 2022 at 6:05 pm

Hi Niels, when running the “kubeadm init” or “kubeadm join” i had this error

[preflight] Running pre-flight checks
[WARNING SystemVerification]: missing optional cgroups: blkio
error execution phase preflight: [preflight] Some fatal errors occurred:
[ERROR FileContent–proc-sys-net-ipv4-ip_forward]: /proc/sys/net/ipv4/ip_forward contents are not set to 1
[preflight] If you know what you are doing, you can make a check non-fatal with `–ignore-preflight-errors=...`

To solve it, i had to set ip_forward content with 1 by following command:

echo 1 > /proc/sys/net/ipv4/ip_forward

Reply



Ramesh
April 2, 2023 at 6:35 am

If the k8s master token expired then also will get this kind of issue.
check using below command to know token expired or not
> kubadm token list
if no results it means token expired

Genrate new token like below
>kubadm token create

Reply



vignesh
September 19, 2022 at 5:31 am

thank you it worked out

Reply



Anees
September 23, 2022 at 10:42 am

Could you add the aws cloud provider configuration into this setup; so that we can utilize ebs persistent storage and AWS ELB.

Reply



vio
October 12, 2022 at 10:52 am

WoW , same here after search aloot on google and try many tutorials , this is finaly work, Good job and many thanks!

Reply



Shanem
October 13, 2022 at 2:59 pm

I currently have Ubuntu 22.04 installed, and I am planning on installing GitLab. I'd want to have a possibly basic Kubernetes environment. I am quite used to kubecti, but I am not confident whether to use MicroK8s, Rancher, MiniKube, or something else. Do you have any opinion about my situation? Thank you so much for answering.

Reply

Dhanesh Malviya
November 1, 2022 at 4:01 pm

Thanks,
This article helped me to successfully setup the kubernetes cluster.

Reply

Steven
November 2, 2022 at 3:43 am

Finally, a tutorial that works! Thanks!!! Other tutorials did not work

Reply

Javier
November 8, 2022 at 12:03 pm

Thank you! At last, one tutorial that works fine!
Again, thanks.

Reply

Sivaraman
November 10, 2022 at 2:47 pm

My last 2 days of installation struggle end up with your blog .. Thanks to your fantastic work...

Reply

Pierre
November 14, 2022 at 5:28 am

Excellent guide. Works perfectly. I had to install AppArmor as it is not installed in lxd VM images.

You used “step 6)” twice.

Reply

Ash Allen
November 15, 2022 at 2:23 pm

This is a great tutorial, thank you for putting it together! I have managed to successfully create a cluster using it, which is awesome! Really appreciate the time you’ve taken here.

Reply

Marcus

November 21, 2022 at 2:31 pm

I don't know why my pods is having crash status on calico

root@k8smaster:~# kubectl get pods -n kube-system -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
calico-kube-controllers-798cc86c47-g95km 1/1 Running 0 6m54s 172.16.16.129 k8smaster
calico-node-2jwsx 1/1 Running 0 6m54s 192.168.60.10 k8smaster
calico-node-5w2bj 0/1 Init:CrashLoopBackOff 5 (2m34s ago) 6m54s 192.168.60.11 k8sworker1
calico-node-8rqx5 0/1 Init:CrashLoopBackOff 6 (22s ago) 6m54s 192.168.60.12 k8sworker2

Reply

prashanth

November 27, 2022 at 7:02 am

Awesome! Its worked after lot of failures, always problem facing in creating pod networks, Thank you so much

Reply

TikAlienware

November 28, 2022 at 9:38 am

Thank you so much, after too many videos, I found this doc very good

Reply

Hugo
December 13, 2022 at 1:44 am

Nice tutorial, you really help me 🙂

Reply

Peter Jaffray
December 20, 2022 at 5:51 pm

Finally – It works as described. After trying to migrate from AWS to baremetal for a few weeks this worked. I really appreciate you!

Reply

DWARA GANESH
December 23, 2022 at 5:51 am

Hi Pradeep,

I have been trying to Set-up the K8S Cluster for the last 10 days. I have watched many youtube videos as well for that. Finally, today I set up the K8S Cluster with the help of your amazing step-to-step guide.

Thank you so much for your efforts.

Reply

Milan
December 28, 2022 at 5:12 pm

Thanks for putting this guide together. Spent hours on various sites following different instructions with no success.
You guide made it nice and easy.

Reply

bhanu
December 28, 2022 at 6:20 pm

Great after 3 days, i am able to set up kubeadm set up using your article.

Reply

andres
January 1, 2023 at 10:01 pm

THE BEST TUTORIAL BY FAR, IT WORKS GRATEFUL, THE LINUX FOUNDATION ONE DIDN'T WORK FOR ME BUT THIS ONE DOES, THANK YOU

Reply

Aswin Hadinata
January 3, 2023 at 2:34 am

Just worked! thank you

so much helped for newbie

Reply

Panchal
January 4, 2023 at 12:43 pm

great article.... after going through so many website this is good one

Reply

Mulyadi Alfatih
January 11, 2023 at 4:19 pm

Hi, nice tutorial, could you please help me, i have been follow the step until install calico, but when i run "$ kubectl get pods -n kube-system", the status appears there are pending and running, see below :

NAME READY STATUS RESTARTS AGE
calico-kube-controllers-7bdbfc669-cwdk5 0/1 Pending 0 8m
calico-node-qk4vp 0/1 Init:ImagePullBackOff 0 8m
calico-node-vbjg8 0/1 Init:ImagePullBackOff 0 8m
coredns-787d4945fb-lzblv 0/1 Pending 0 51m
coredns-787d4945fb-rqnkc 0/1 Pending 0 51m
etcd-k8smaster.alfatih.v2 1/1 Running 0 51m
kube-apiserver-k8smaster.alfatih.v2 1/1 Running 0 51m

kube-controller-manager-k8smaster.alfatih.v2 1/1 Running 0 51m
kube-proxy-52r46 1/1 Running 0 11m
kube-proxy-w5hdz 1/1 Running 0 51m
kube-scheduler-k8smaster.alfatih.v2 1/1 Running 0 51m
==========
==
FYI, i run this K8S on virtual box.

Reply

mattias
January 13, 2023 at 3:55 am

great guide ! easy to understand and every command worked like it should. big thanks

Reply

Tater-Tots
January 13, 2023 at 10:57 pm

I am running into an issue with the nginx-app is not being exposed properly. There is no “External-IP”.

kubectl get svc nginx-app
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
nginx-app NodePort 10.109.94.99 80:30817/TCP 6m42s

Can anyone assit?

Reply

Nathan
January 15, 2023 at 1:41 pm

Thanks Pradeep for this wonderful document each and everything well explained and documented.
Much Appreciated for your efforts.

Reply

Abhiranjan
January 24, 2023 at 7:12 pm

Hey pradeep, thanks for this document . it's really help me to setup the cluster. all other doc make me sick .

Reply

Allan Turnbull
January 27, 2023 at 2:48 pm

Thanks for the step by step guide, clear and concise.

Thanks for taking to time to post the article.

It was my second attempt at installing K8s and the second post I looked at, which is quite good considering the amount of posts out there.

Now to get the dashboard installed and setup for using kvm instead of containers.

Reply

Yamil ortega
February 15, 2023 at 3:01 am

excellent tutorial, Last week works perfectly, but today, something happened with the calico step.

$ curl 'https://projectcalico.docs.tigera.io/manifests/calico.yaml' -O
$ kubectl apply -f calico.yaml
can you help?

Reply

Pradeep Kumar
February 15, 2023 at 6:17 am

Hi,

Use following command to install calico,

$ kubectl apply -f 'https://raw.githubusercontent.com/projectcalico/calico/v3.25.0/manifests/calico.yaml'

I have updated the same in post too.

Reply

Yamil Ortega
February 15, 2023 at 5:09 pm

works like a charm!! Thank you! Cheers from Mexico!

Reply

moko
February 19, 2023 at 2:33 am

hi, after running kubeadm init, i ran kubectl cluster-info and it worked.

but after a few minutes, i ran again kubectl cluster-info, error occured
“k8smaster.example.net:6443 was refused – did you specify the right host or port?”

can you help me? i have followed all instruction in this tutorial.

i’m using ubuntu 22.04, running on EC2 Instance (AWS)

Reply

Foo
February 27, 2023 at 7:51 pm

Try running
sudo swapoff -a && sudo sed -i ‘/ swap / s/^\(.*\)$/#\1/g’ /etc/fstab

Reply

KP
February 28, 2023 at 3:11 pm

Can anyone help me with error.

ubuntu@k8smaster:~$ sudo kubeadm init –control-plane-endpoint=k8smaster.example.net

[init] Using Kubernetes version: v1.26.1
[preflight] Running pre-flight checks
error execution phase preflight: [preflight] Some fatal errors occurred:
[ERROR Port-6443]: Port 6443 is in use
[ERROR FileAvailable–etc-kubernetes-manifests-kube-apiserver.yaml]: /etc/kubernetes/manifests/kube-apiserver.yaml already exists
[ERROR FileAvailable–etc-kubernetes-manifests-kube-controller-manager.yaml]: /etc/kubernetes/manifests/kube-controller-manager.yaml already exists
[ERROR FileAvailable–etc-kubernetes-manifests-kube-scheduler.yaml]: /etc/kubernetes/manifests/kube-scheduler.yaml already exists
[ERROR FileAvailable–etc-kubernetes-manifests-etcd.yaml]: /etc/kubernetes/manifests/etcd.yaml already exists
[ERROR Port-10250]: Port 10250 is in use
[ERROR Port-2379]: Port 2379 is in use
[ERROR Port-2380]: Port 2380 is in use
[ERROR DirAvailable–var-lib-etcd]: /var/lib/etcd is not empty
[preflight] If you know what you are doing, you can make a check non-fatal with `–ignore-preflight-errors=...`
To see the stack trace of this error execute with –v=5 or higher

Reply

Pradeep Kumar
March 1, 2023 at 1:57 am

It seems like, you already have tried installing k8s cluster on your Ubuntu system because of that port 6443 is already in use.

Reply

Pradeep Kumar
March 3, 2023 at 7:29 pm

$ sudo kubeadm reset cleanup-node
and then try init again

Reply

Derrick
March 4, 2023 at 4:43 pm

Hello,
I get this error on the worker nodes when I run `sudo kubeadm join`

[failure loading certificate for CA: couldn't load the certificate file /etc/kubernetes/pki/ca.crt: open /etc/kubernetes/pki/ca.crt: no such file or directory, failure loading key for service account: couldn't load the private key file /etc/kubernetes/pki/sa.key: open /etc/kubernetes/pki/sa.key: no such file or directory, failure loading certificate for front-proxy CA: couldn't load the certificate file /etc/kubernetes/pki/front-proxy-ca.crt: open /etc/kubernetes/pki/front-proxy-ca.crt: no such file or directory, failure loading certificate for etcd CA: couldn't load the certificate file /etc/kubernetes/pki/etcd/ca.crt: open /etc/kubernetes/pki/etcd/ca.crt: no such file or directory]

What can I do to resolve this?

Reply

Norm
March 12, 2023 at 3:19 am

I had this problem too, then I realized the kubeadm init command output listed two separate join commands. The first was to add control-plane nodes, and that was the command I'd copied. Make sure you get the second command, which does not have the "–control-plane" parameter

Reply

Victor Moreno
March 11, 2023 at 7:07 am

Thanks very much.
Just executed the steps are in there, and all worked great!!! thanks

Reply

Lu
March 12, 2023 at 5:08 pm

Excellent, this saved me days of time trying to fix coredns issue for ubuntu18.04.

Reply

Saeed
March 29, 2023 at 9:27 pm

I wasted a few days and watched half a dozen videos – until I found your plain solution.

Thanks

Reply

Bish
March 30, 2023 at 11:19 pm

Great Job Pradeep –

BTW, in this guide – I don't see a reference of "sudo kubeadm init –pod-network-cidr=192.168.0.0/16" command anywhere. How POD Network is configured, what would be the POD Network CIDR (if anything default)? and where to find it? Can we update the config later?
Thanks
~Bish

Reply

Tom Dee
April 6, 2023 at 3:11 pm

Hi == agreed best step by step –others failed.. Issuue at the end though.
NO ENDPOINTS Here :

~$ kubectl describe svc nginx-app
_____
Name: nginx-app
Namespace: default
Labels: app=nginx-app
Annotations:
Selector: app=nginx-app
Type: NodePort
IP Family Policy: SingleStack
IP Families: IPv4
IP: 10.99.225.11
IPs: 10.99.225.11
Port: 80/TCP
TargetPort: 80/TCP
NodePort: 32495/TCP

Endpoints:
Session Affinity: None
External Traffic Policy: Cluster
Events:
————————————————- and then
curl ‘http://worker1 ip address:31246’
curl: (7) Failed to connect to 10.165.2.129 port 31246 after 0 ms: Connection refused ??
Thanks

Reply

Pradeep Kumar
April 6, 2023 at 3:52 pm

Hi Tom,

As per your nginx service, nodeport is 32495 but in your curl command you are using different port. Please crosscheck the nodeport.

Reply

Tom Dee
April 6, 2023 at 4:16 pm

Never Mind.. I changed the port to 32495 — all good!!

Reply

Suman Bhandari
April 8, 2023 at 7:50 pm

Thanks for the guide.
Seems like I too have same story of others. Tried several docs and videos. Finally this helped me.

Reply

Suman Bhandari
April 9, 2023 at 7:36 am

After following your steps I was able to install kubeadm successfully. I tried so many docs, videos for a week finally got to this so far.

But now I get the error like this again. This was the reason i tried several installation guide. What could be the reason behind this error ?

pseudouser@masterkube:~$ kubectl get nodes
E0409 13:16:54.278679 8776 memcache.go:265] couldn't get current server API group list: Get "http://localhost:8080/api?timeout=32s": dial tcp 127.0.0.1:8080: connect: connection refused
E0409 13:16:54.278888 8776 memcache.go:265] couldn't get current server API group list: Get "http://localhost:8080/api?timeout=32s": dial tcp 127.0.0.1:8080: connect: connection refused
E0409 13:16:54.279978 8776 memcache.go:265] couldn't get current server API group list: Get "http://localhost:8080/api?timeout=32s": dial tcp 127.0.0.1:8080: connect: connection refused
E0409 13:16:54.281517 8776 memcache.go:265] couldn't get current server API group list: Get "http://localhost:8080/api?timeout=32s": dial tcp 127.0.0.1:8080: connect: connection refused
E0409 13:16:54.282856 8776 memcache.go:265] couldn't get current server API group list: Get "http://localhost:8080/api?timeout=32s": dial tcp 127.0.0.1:8080: connect: connection refused
The connection to the server localhost:8080 was refused – did you specify the right host or port?
pseudouser@masterkube:~$

Reply

Salman Malik
May 8, 2023 at 9:00 am

Hi Suman,

Please follow all the process mentioned on step number 6.

Reply

Anand
May 18, 2023 at 12:22 am

Thanks a lot, I have spent 1 weeks, was getting coredns-pending in centos and coredns-creating in ubuntu with docker and containerd setup. Finally, it works with your perfect details.

Reply

vinay
June 1, 2023 at 3:36 pm

After search lots on google and try many tutorials , this is finally work, Good job and many thanks!

Reply

Michael
July 14, 2023 at 11:30 am

This material is a life saver! So on-point. Thank you for sharing

Reply

Joel
July 15, 2023 at 6:27 am

Hi Niels and thank you for this very important document. It is by far the best I came across online.

I am having issues with the worker nodes joining the cluster.
Here is my setup on VMWare workstation. The VMs are all from CentOS8.
192.168.234.130 k8smaster.example.net k8smaster
192.168.234.131 k8sworker1.example.net k8sworker1
192.168.234.132 k8sworker2.example.net k8sworker2

Here is the error I am getting on both worker nodes:

devops@k8smaster:~$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
k8smaster.example.net NotReady control-plane 2d7h v1.27.3
devops@k8smaster:~$

devops@k8sworker1:~$ sudo kubeadm join k8smaster.example.net:6443 –token vt4ua6.wcma2y8pl4menxh2 –discovery-token-ca-cert-hash sha256:0494aa7fc6ced8f8e7b20137ec0c5d2699dc5f8e616656932ff9173c94962a36
[preflight] Running pre-flight checks
error execution phase preflight: couldn’t validate the identity of the API Server: could not find a JWS signature in the cluster-info ConfigMap for token ID “vt4ua6”
To see the stack trace of this error execute with –v=5 or higher
devops@k8sworker1:~$

devops@k8sworker2:~$ sudo kubeadm join k8smaster.example.net:6443 –token vt4ua6.wcma2y8pl4menxh2 –discovery-token-ca-cert-hash sha256:0494aa7fc6ced8f8e7b20137ec0c5d2699dc5f8e616656932ff9173c94962a36
[sudo] password for devops:
[preflight] Running pre-flight checks
error execution phase preflight: couldn't validate the identity of the API Server: could not find a JWS signature in the cluster-info ConfigMap for token ID "vt4ua6"
To see the stack trace of this error execute with –v=5 or higher
devops@k8sworker2:~$

I processed with the CNI installation in hope that it will hep. But this is what I got below. The worker nodes are still not joining the cluster.

devops@k8smaster:~$ kubectl get pods -n kube-system
NAME READY STATUS RESTARTS AGE
calico-kube-controllers-6c99c8747f-2skjx 1/1 Running 0 64s
calico-node-v498j 1/1 Running 0 64s
coredns-5d78c9869d-5ljmp 1/1 Running 0 2d7h
coredns-5d78c9869d-qn29s 1/1 Running 0 2d7h
etcd-k8smaster.example.net 1/1 Running 2 (35m ago) 2d7h
kube-apiserver-k8smaster.example.net 1/1 Running 2 (35m ago) 2d7h
kube-controller-manager-k8smaster.example.net 1/1 Running 2 (35m ago) 2d7h
kube-proxy-vzw5z 1/1 Running 2 (35m ago) 2d7h
kube-scheduler-k8smaster.example.net 1/1 Running 2 (35m ago) 2d7h
devops@k8smaster:~$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
k8smaster.example.net Ready control-plane 2d7h v1.27.3
devops@k8smaster:~$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
k8smaster.example.net Ready control-plane 2d7h v1.27.3

Could you please help? Many Thank in advance.

Joel

Reply

Minh Dat
August 2, 2023 at 4:07 am

you can use command “kubeadm token create –print-join-command” to get command to join
Note: add sudo before command

Reply

Somuhd
August 4, 2023 at 10:28 pm

After lots of efforts to setup by watching you tube video’s and blogs finally found solution and I able setup cluster in 10 minutes, If follow all the steps it will be successful in first attempt.

Thank you very much for such wonderful information,

Reply

Nizar ibrahim
August 9, 2023 at 6:10 pm

This is fantastic bro.since three weeks iam spending my time to get righteous one .finally its done

Reply

Julio
August 25, 2023 at 2:01 pm

Thank you so much! As others already wrote this is the working solution after trying lots of other tutorials. Would be nice to get deeper into network for example where the NGINX Service got the IP from and so on.

Reply

Amit Karia
August 28, 2023 at 10:05 am

thank you so much for this great article, for the first time I have been able to setup a Kubernetes cluster at home.

Reply

Vinita
August 31, 2023 at 5:09 am

Great tutorial...I was trying to do it from 10 days and now got it done in 2 hours... Thank you so much

Reply

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